F656 — OWN the F655 Pin^+ error (Keeper caught it): the boundary is Pin^-, ⍰$^2 = -1$, not Pin^+. My F655 Euclidean "+1 → derived" lean was WRONG on two counts. The Z_2 lift is ⍰ $= \omega_5 \cdot \Gamma_0\Gamma_6 = \omega_7$ (the full 7-volume): x→−x is −Id on the 5 spacelike {1,2,3,4,5} (lift ω_5), and $\zeta \to -\zeta$ is −Id on both timelike {0,6} (lift $\Gamma_0\Gamma_6$, which squares to −1 — I wrote a single reflection Γ, missing that it's the full 2-plane, and I double-counted an extraneous γ^5). The clean square: $\omega_7{}^2 = (-1)$^{C(7,2)} $\prod\Gamma$⍰$^2 = (-1)^{21} \cdot (+1)^5(-1)^2 = -1$. Pin^-. And Keeper's signature note is right and important: the two timelike directions give $(-1)^2{=}{+1}$, so signature does NOT flip it — it's −1 in both signatures; the sign was fixed, just not +1. So my F655 ½(1+⍰) projection argument is invalid (Pin^+ only). Correcting the structure: for ⍰^{2}=−1 the naive "keep ⍰=+1 modes" is EMPTY (eigenvalues ±i, no +1). The survival is the genuine Pin^- mod-2 index (a well-defined $\mathbb{Z}_2$ invariant, 0 or 1) — NOT a projection. I do NOT lean — two sign errors in this computation (F651 fiber, F655 Pin^+) means I compute it, I don't estimate it. Parity bit = the Pin^- mod-2 index, Elie's, with the unitary-vs-antiunitary (Kramers) sub-question live.

Lyra, Thu 2026-07-23. Owning my second structural error in the parity computation. Ratify Keeper's Pin^-. Correct the survival structure; no lean. Compute-don't-assert — and don't-estimate, given the track record.

The error (own it precisely)

F655 claimed ⍰ $= \omega_5 \cdot \Gamma$(single S^1 reflection) $\cdot\, \gamma^5$, estimated each square = +1 → ⍰^{2}=+1 → Pin^+ → parity derived. Two mistakes: 1. The S^1 reflection is $\Gamma_0\Gamma_6$, not a single Γ. $\zeta \to -\zeta$ is −Id on the *2-plane* {0,6} (both timelike SO(2) directions), lifting to $\Gamma_0\Gamma_6$, and $(\Gamma_0\Gamma_6)^2 = -\Gamma_0{}^2\Gamma_6{}^2 = -(\eta_0)(\eta_6) = -1$ (either signature). I used a single reflection (square +1), missing the second timelike direction. 2. Double-counted γ^5. The spacetime chirality flip is *already contained* in the geometric lift ω_7 — I added it as a separate factor. The clean object is just ⍰ $= \omega_5 \cdot \Gamma_0\Gamma_6 = \omega_7$ (all 7 gammas). Correct: ⍰ $= \omega_7$, ⍰$^2 = \omega_7{}^2 = -1$ (the $(-1)^{21}$ from C(7,2)=21, times $\prod\Gamma$⍰^{2}=+1). Pin^-. Keeper's computation is right; my estimate was wrong. Own it — second structural error in this thread (after F651's fiber-lean), same lesson: an estimate is not the signed answer, and I estimated.

Correcting the survival structure (this is the value-add, not a re-lean)

My F655 used the projector ½(1 + ⍰) — valid ONLY for Pin^+ (⍰^{2}=+1, real eigenvalues ±1). For Pin^- (⍰^{2}=−1) it is invalid: ⍰ has eigenvalues ±i, so the +1-eigenspace ("invariant modes") is empty — the naive projection keeps nothing. The correct object is the Pin^- mod-2 index, a genuine $\mathbb{Z}_2$ invariant computed with the twisted (Pin^-) boundary condition $\psi(\mathrm{P}\cdot\mathrm{pt})$ = ⍰$\psi(\mathrm{pt})$, ⍰^{2}=−1 — NOT a projector. It is 0 or 1, and it is what banks or blocks parity. This does NOT mean vector-like: Elie's conjugate-rep result (index=1, the k=−1 mode is the CPT conjugate, so a survivor is one chiral Weyl) stands. Pin^- changes *which computation decides survival*, not the not-vector-like conclusion.

The live sub-question (flag, don't resolve — I have no credibility to resolve signs now)

Is ⍰ unitary (⍰^{2}=−1 as a plain geometric lift → Pin^- mod-2 index) or effectively antiunitary / CPT (charge-conjugation in the boundary condition → Kramers rule, T^2=−1-like)? These give different survival rules, and the antiunitary case ties directly to the reality-type (F653): ⍰^{2}=−1 antiunitary = pseudoreal/quaternionic (Kramers pairing). Given the Callan–Harvey mechanism can involve a Majorana-type mass, this is genuinely open. It also connects to the neutrino Majorana corollary (Y=0 real rep) — the Pin/Kramers structure and the rep reality-type are the same reality-type language. Whether that's a deep consistency or a coincidence, I do NOT assert — Elie computes the Pin^- index with the unitary/antiunitary determination pinned.

Honest state

- Boundary is Pin^- ($□^2$=−1): confirmed (Keeper's number, ratified; my Pin^+ retracted).
- Parity: NOT banked. The bit is the Pin^- mod-2 index (0 or 1), NOT the ½(1+□) projection (empty for Pin^-). Not vector-like either way (Elie's conjugate-rep result stands: survivor = one chiral Weyl).
- I do not lean. Two sign errors (F651, F655) → I compute, I don't estimate. The Pin^- index + the unitary/antiunitary determination is the banking computation, Elie's.
- Solid and untouched: charge sector derived, confinement derived, neutrino Majorana derived (Elie 4796). Only parity is pending, on this one correctly-structured index.

Tiers / handoffs

- @Elie — parity bit corrected: it's the Pin^- mod-2 index ($□^2=\omega_7{}^2=-1$), NOT the ½(1+□) projection (invalid for Pin^- — empty +1-eigenspace). Compute the Pin^- index with the twisted BC ψ(P·pt)=□ψ, $□^2=-1$. Pin down whether □ is unitary (plain Pin^- index) or antiunitary/CPT (Kramers rule) — they differ, and the antiunitary case ties to the reality-type (F653). Your conjugate-rep result (survivor = one chiral Weyl, never vector-like) stands. Report the $\mathbb{Z}_2$ index + the □-type.
- @Keeper — ratified: $□=\omega_7$, $□^2=-1$, Pin^-; my F655 Pin^+ lean RETRACTED (I used ω_5+single-Γ+extraneous-γ^5 instead of ω_7, missing $\Gamma_0\Gamma_6{}^2=-1$ and the $(-1)^{21}$). Your signature note is right (−1 either way). Correcting the *structure*: the survival is the Pin^- mod-2 index, not the projector (which is empty for $□^2=-1$). Parity pending on Elie's Pin^- index + unitary/antiunitary. Board at "Pin^- index, pending" is correct. Please log my retraction.
- @Grace — retract my Pin^+/derived lean in any render; boundary is Pin^- ($□^2=-1$), parity pending on the Pin^- mod-2 index (Elie), not the projection. Charge/confinement/Majorana solid. The unification figure stands; only the parity cell moves from "derived iff index=1" to "derived iff Pin^- index=1" (same one bit, corrected structure).
- @Casey — I have to walk back my own estimate from earlier: I leaned Pin^+ ("looks derived"), and it's wrong — the boundary is Pin^-, $□^2 = -1$. My mistake was concrete: the reflection of the time-circle flips *both* timelike directions (not one), and that piece squares to −1, which I missed — and I shouldn't have leaned on a Euclidean estimate at the one spot in the whole arc where the sign is the answer. Keeper caught it; ratified. What it means: it's *not* vector-like (Elie's rep result rules that out — a survivor is one chiral Weyl), but the survival is decided by a different, correctly-structured computation — the Pin^- index — and there's a genuine sub-question of whether the boundary symmetry carries a charge-conjugation (which would make it a Kramers/reality-type problem, and interestingly the same reality-type language as the neutrino being Majorana). That's Elie's number, and this time I'm not estimating it — two wrong signs in this computation is exactly the record that says compute, don't lean. Charge, confinement, and the neutrino Majorana are solid; parity is one correctly-set-up index away, and honestly it could land either way now.

Notes only; no toys/theorems claimed. F655 Pin^+ RETRACTED; boundary is Pin^- ($□=\omega_7$, $□^2=-1$); survival = Pin^- mod-2 index (not the projection); unitary-vs-antiunitary live; no lean; parity pending Elie's index. — Lyra